Evolutionary Algorithm with Anisotropic Mutation Operator and Proven Global Convergence

***Abstract*—The paper proposes a novel evolutionary numerical algorithm for global optimization, using anisotropic mutations with a shift in their distribution centers and the choice of a parent agent based on the Lotka-Volterra model. The main objective of the paper is to develop a numerical algorithm for generating a test point sequence that converges to the global maximum (in probability). The main advantage of the proposed algorithm is the mathematically proven convergence of the test point sequence to the global extremum with probability one. Two theorems are proved in the paper: the first states that any open subset in the search space contains at least one test point with probability one; the second states that the test point sequence will concentrate over time in an arbitrarily small neighborhood of the global extremum point. We compared the performance of this algorithm with several popular metaheuristic algorithms on one hundred randomly generated non-differentiable relevant test functions. The comparison showed the advantage of the accuracy and convergence rate of the proposed algorithm. As a result, the developed algorithm was applied to solve the main problem – finding the optimal control of the UAV when turning in conditions of turbulent gusts of wind. The experiments showed the applicability and efficiency of our algorithm in solving technical problems of high-complexity optimal control.**

***Index Terms*—global optimization, evolutionary algorithm, algorithm convergence, anisotropic mutations, optimal control, direct methods of optimal control.**

I. Introduction

OPTIMAL control problems are pivotal in technical engineering design, for example, for the construction of unmanned aerial vehicles [1, 2].

Most applied optimal control problems require the use of numerical solution methods. Indirect methods of this solution are developed on the basis of the necessary optimality conditions in the form of the traditional maximum principle [3–6]. However, recently direct solution methods have become increasingly popular. They are reduced to finding the best value for the function of a finite number of variables [7]. Direct solution methods make it possible to bypass difficulties of the traditional approach [6]. They are especially relevant when the maximum principle does not provide sufficient information to construct a solution [8, 9]. The growing popularity of direct methods is explained by the growth in the power of computing technology and the development of productive numerical algorithms for optimizing functions in multidimensional spaces. Such algorithms include metaheuristic global optimization algorithms, which are effectively used to solve optimal control problems in engineering design [10, 11]. Metaheuristic algorithms imitate some natural or social processes [12–17]. One of the most popular methods of this class is differential evolution and its numerous modifications - evolutionary methods based on the ideas of biological evolution [18–22]. At the same time, there is no best algorithm for all problems at once [23]. On average, methods that show high performance on test examples sometimes work poorly on a real practical problem and get stuck at points of false local extremum. For each applied problem, it is necessary to select a method that best suits its features. In this regard, the arsenal of methods is constantly expanding, and new improved modifications of known algorithms appear [24].

The general drawback of metaheuristic algorithms is the need for guaranteed convergence of the algorithm to the point of a global solution in any strict mathematical sense. It is impossible to guarantee that the algorithm will never fall into a false local extremum trap [25, 26]. In this regard, a situation is possible when different algorithms give different solutions to the same problem [27]. Obviously, in applied problems of optimal control, such as choosing UAV settings, it is essential to exclude the possibility of false decisions that lead to additional losses. Thus, the problem of developing an algorithm with justified convergence to a global optimum that is equal in performance to existing advanced evolutionary algorithms remains relevant [28].

Recently, a new evolutionary algorithm SoFA (Survival of the Fittest Algorithm) was developed [29], which generates a sequence of test points. The test point is a point in the search area at which the value of the objective function is calculated when implementing the algorithm. The researchers proved the convergence of a sequence of test points in probability to the global maximum point with probability one. Later, the performance of the algorithm was significantly increased [30]. Then a new improved modification SoFDE was created on the base of a Cauchy distribution [31]. SoFA has shown its efficiency in solving a number of applied problems [32–34]. However, numerical experiments of optimal aircraft control problems show that it is less productive than other evolutionary algorithms due to a large number of additional nonlinear constraints on the stability of the solution and the duration of transient processes during stabilization.

This study is devoted to developing a new evolutionary algorithm based on SoFA that guarantees convergence to the global optimum point and sufficient performance for optimal unmanned aerial vehicle settings control. The contributions of this article are as follow:

1) Unlike traditional metaheuristic algorithms, which rely on empirical results, the proposed algorithm provides a rigorous theoretical proof of convergence, addressing the shortcomings of empirical approaches.
2) The algorithm achieves faster and more accurate optimization, offering significant improvements in efficiency and practicality over existing algorithms. The acceleration of the algorithm is achieved by using a new idea of shifts in the distribution centers of mutations with a monotonic

decrease in the shift parameter depending on the number of steps and a change in the population size in accordance with the Volterra-Lotka model.

3) The proposed algorithm allows us to effectively construct an optimal control, including situations where the maximum principle does not provide sufficient information to find the solution, and to overcome the limitations of existing algorithms for numerically solving the problem of controlling the UAV autopilot.

The paper is organized as follows. In Section II, we give some preliminary concepts; in Section III, we present the newly developed algorithm. In Section IV, we provide proof of theorems asserting the convergence of the algorithm to the global optimum. Section V describes computational experiments comparing the efficiency of the proposed algorithm with similar ones and an experiment to determine the optimal control of a light UAV using our algorithm. A discussion of the obtained results is given in Section VI. Finally, in Section VII, we formulate a conclusion about the efficiency of the new evolutionary algorithm and describe prospects for further research.

II. Preliminary concepts

A. Statement of Problem

In this study, we address an optimization problem involving a continuous positive function. Let $F(z)$ represent a continuous positive function defined within the hyperparallelepiped Π, where Π is the set of points $z = (z_1, \ldots, z_n)$ in n-dimensional Euclidean space, the coordinates z_i of these points satisfy conditions $a_i^L \leq z_i \leq a_i^R, i = \overline{1,n}$. We denote $z' = (z'_1, \ldots, z'_n)$ as the point where $F(z)$ attains its maximum value within the defined region.

$$F(z') = \max_{z \in \Pi} F(z). \tag{1}$$

The objective is to develop a numerical algorithm for generating a test point sequence $z^k = (z_1^k, ..., z_n^k)$ that converges to the global maximum z' (in probability).

Subsequently, the rate of convergence in approximation accuracy should be compared with those of established global optimization algorithms on standard test functions. Additionally, the feasibility of applying this procedure to practical problems in the optimal control of UAV systems needs to be evaluated.

B. Features of the new algorithm

This paper introduces SoFAM (Survival of the Fittest Anisotropic Mutations) algorithm, a novel evolutionary algorithm based on the SoFA approach. Compared to the basic SoFA, the new SoFAM algorithm uses several new fundamental ideas. Here, the population size changes to the dynamics of the Volterra-Lotka model. The Volterra-Lotka difference model of coexisting species dynamics and the idea of using it to construct an evolutionary optimization algorithm were discussed in detail in [35]. A brief commentary on the relationship of this model with the modernization of the SoFA optimization algorithm is also given in [29]. The mutation operator acts differently than the SoFA, using Cauchy distribution and shifting the distribution center (location parameter) like the mutation operator in differential evolution. But unlike the latter, the shift parameter (regulating its length) gradually decreases with an increase in the number of steps. It accelerates the approach to the global maximum point at the final stages of the process. The proposed SoFAM algorithm has also significant differences from the differential evolution. First, this difference lies in choosing the reference (parent) point for forming the next mutation. According to the differential evolution, this point is selected from the population points with equal probability. In the SoFAM algorithm, this selection is also random, but the distribution of selection probabilities corresponds to the dynamic processes of competition and selection in the Volterra-Lotka model. Instead of the crossover operator, typical for evolutionary algorithms, here, like in the SoFA, mutations of two types are used - strong and weak. In addition, we can see that performing the crossover operation in differential evolution is equivalent to only shifting the center of each gene's mutation distribution (location parameter) in the SoFAM algorithm. In the SoFAM algorithm, after performing this transformation, the Cauchy distribution uses an additional random scatter relative to the shifted center (location), which has no analog in differential evolution. This novelty provides an additional examination of the search space and positively affects productivity. The intensity of this scatter (scale parameter) is not constant but gradually decreases to zero, so the convergence of the proposed algorithm is achieved.

C. Definition of pre-set parameters and control functions

In the following algorithm, each test point is referred to as an "agent", and each component of the test point is termed a "gene".

The SoFAM algorithm employs several key control functions and parameters: $\zeta(k)$ governs mutation intensity, is defined as a positive numerical sequence that monotonically decreases to zero as the step number k increases indefinitely. The function $\phi(k)$ regulates agent selection for subsequent mutations, increasing monotonically with k and satisfying the condition: $\phi(k) \to \infty$ as $k \to \infty$. The function $H(k)$ with values in the interval (0, 1) controls the shift in the distribution center during mutation; it monotonically decreases to zero as the step number k increases indefinitely. The parameter G, a constant within the interval (0, 1), serves as a threshold for determining whether a mutation is classified as strong or weak; the parameter γ, a constant within the interval (0, 1), serves as a threshold for including an agent in the population. Finally, N represents the interval between strong and weak mutations and is chosen as a natural number comparable to the total number of steps k.

D. Variable notations

The following variables are used in the SoFAM algorithm: agent z^k is a point of Π with coordinates $(z_1^k, ..., z_n^k)$; m is size (number of agents) of the initial population. $\mu(i,k)$ is the indicator that shows whether the i-th agent is included in the population at step k $(k > m)$. This indicator is set to

1 if the i-th agent is included at that step and 0 otherwise. The reference agent $w^k = (w_1^k, ..., w_n^k)$ and two additional agents $v^k = (v_1^k, ..., v_n^k)$, $u^k = (u_1^k, ..., u_n^k)$ are elements of the population used for the mutation operator at the k-th step. p_j^k is the probability of selecting the j-th agent of the population as a reference or additional agent at the k-th step. σ_{ik} is the scale parameter, m_i^k is the shift parameter of Cauchy distribution for the mutation of the i-th gene at the k-th step. B_{ik} is the normalized constant for Cauchy distribution.

III. ALGORITHM

A. Preparatory stage of the algorithm: initial population

Initially, m points (agents) $z^1, ..., z^m$ are randomly and uniformly selected from the hyperparallelepiped Π. The function values $F(z^1), ..., F(z^m)$ are calculated for these agents. These agents constitute the initial population. Then we set $\mu(i, m+1) = 1$ for $i = \overline{1, m}$.

B. Main stages of the algorithm: population evolution

At each k-th step beyond the initial m steps ($k > m$), a new agent $z^k = (z_1^k, ..., z_n^k)$ is selected in Π through a series of operations. The first operation involves identifying the agent $z^{k*} = (z_1^{k*}, ..., z_n^{k*})$ with the highest objective function value F within the population. This agent is determined by the following equation:

$$F(z^{k*}) = \max_{j=\overline{1,k-1}} \Big(\mu(j,k) \cdot F(z^j)\Big). \tag{2}$$

Next, a reference agent w^k is randomly selected from the population with the selection probability for each agent determined by the following probability distribution:

$$p_j^k = \frac{\big(F(z^j)\big)^{\phi(k)}}{\sum_{l=1}^{k-1} \Big(\mu(l,k)\Big(F(z^l)\Big)^{\phi(k)}\Big)}. \tag{3}$$

In addition to the reference agent, two more agents, v^k and u^k, are selected from the population using the same probability distribution. Formula (3) corresponds to the dynamics of the normalized number of competing species in a community that satisfies the Volterra-Lotka hypotheses. In this case, the value $F(z^j)$ plays the role of the fitness of the j-th species – the reproduction coefficient in favorable conditions.

Once the agents are selected, an anisotropic mutation is applied to each gene (coordinate) of the agent. The mutation type, either strong or weak, is determined by comparing a uniformly distributed random variable ξ_i^k (drawn from the interval [0, 1]) with the parameter G. If $\xi_i^k < G$, a strong mutation is applied to the ith gene; otherwise, a weak mutation is performed.

After determining the mutation type, the scale parameter σ_{ik} and the shift parameter m_i^k of the Cauchy distribution are calculated for the mutation of each i-th gene.

- for the strong mutation:

$$m_i^k = w_i^k + H(k)\big(z_i^{k*} - w_i^k\big) + H(k)\big(v_i^k - u_i^k\big); \tag{4}$$

$$\sigma_{ik} = \zeta(k); \tag{5}$$

- for the weak mutation:

$$m_i^k = w_i^k; \tag{6}$$

$$\sigma_{ik} = \zeta(k+N). \tag{7}$$

Each gene z_i^k of the new agent is randomly selected from a Cauchy distribution defined over the interval $[a_i^L, a_i^R]$, with the probability density function:

$$f_{cauchy}(z_i^k) = B_{ik} \frac{\sigma_{ik}}{(z_i^k - m_i^k)^2 + \sigma_{ik}^2}, \tag{8}$$

the constant B_{ik} is determined based on the normalization condition:

$$B_{ik} = \left(\int_{a_i^L}^{a_i^R} \frac{\sigma_{ik}}{(z_i^k - m_i^k)^2 + \sigma_{ik}^2} dz_i^k \right)^{-1}. \tag{9}$$

After generating the new agent z^k, its objective function value $F(z^k)$ is computed. Then the population for the next $k+1$ step is formed according to the following rule:

$$\mu(j, k+1) = \begin{cases} 1, & \text{if } \dfrac{\big(F(z^j)\big)^{\phi(k)}}{\sum\limits_{l=1}^{k} \big(F(z^l)\big)^{\phi(k)}} > \gamma, \quad j = \overline{1,k} \\ 0, & \text{otherwise} \end{cases}.$$

The process is then repeated for subsequent steps.

C. Final stage of the algorithm: obtaining the result

In the simplest case, the algorithm stops when a specified number of steps is exhausted.

The solution to the problem is then determined by identifying the best point, denoted as j^*, where:

$$j^* = arg \max_j F(z^j),$$

the optimal point is denoted as z^{j^*}, with $F(z^{j^*})$ representing the value of the objective function at this point.

D. Pseudocode of the SoFAM algorithm

Here we present a pseudocode of the optimisation algorithm SoFAM (Algorithm 1). Algorithm 1 applies some axillary operators, which are listed below as Algorithms 2, 3, 4 and 5.

Algorithm 1. SoFAM

Require: $n, \Pi \leftarrow \{[a_i^L, a_i^R], i = \overline{1,n}\}, m, Q, N, \gamma,$
$number_of_steps, Objective_Fun;$
$population \leftarrow Generator(n, \Pi, m);$
$population_F \leftarrow Calculator_F(population,$
$Objective_Fun);$
$NP = m;$
for $k := m+1 \rightarrow number_of_steps$ **do**
  $(reference_point, reference_index) \leftarrow Selector(k,$
  $population, population_F, NP);$
  $(shift_point_1, shift_index_1) \leftarrow Selector(k,$
  $population, population_F, NP);$
  $(shift_point_2, shift_index_2) \leftarrow Selector(k,$

```
    population, population_F, NP);
    max_index ← Find_Index_of_Max(population_F);
    max_population ← population[max_index];
    mutation ← Mutation_Operator(k, Π, Q, N,
    reference_point, shift_point_1, shift_point_2,
    max_population);
    NP := NP + 1;
    populaton[NP] ← mutaition;
    population_F[NP] ← Objective_Fun(NP);
    SumF := 0;
    for l := 1 → NP do
      SumF := SumF + (Objective_Fun(l))^{φ(k)};
    end for
    pi = 1;
    for j := 1 → NP do
      if ((popilation_F(j))^{φ(k)}/SumF ≥ γ then
        population[pi] ← populatin[j];
        population_F[pi] ← populatin_F[j];
        pi := pi + 1;
      endif
    end for
    NP := pi − 1
  end for
  max_index ← Find_Index_of_Max(population_F);
  return population[max_index];
```

Algorithm 2. Generater

```
  Require: n, Π, m;
  for i := 1 → m do
    for j := 1 → n do
      current_point[j] ← rand(a_i^L, a_i^R);
    end for population[i] ← current_point;
  end for
  return population;
```

Algorithm 3. Calculater_F

```
  Require: population, Objective_Fun;
  for i := 1 → m do
    population_F[i] ← Objective_Fun(population[i]);
  end for
  return population_F;
```

Algorithm 4. Selector

```
  Require: k, population, population_F, NP;
  for i := 1 → NP do
    probability[i] ← (population_F[i])^{φ(k)};
  end for
  max ← probability[1];
  for i := 1 → NP do
    if probability[i] > max then
      max ← probability[i];
    end if
  end for
  sum ← 0;
  for i := 1 → NP do
    probability[i] ← probability[i]/sum;
    sum ← sum + probability[i];
  end for
  for i := 1 → m do
    probability[i] ← probability[i]/sum;
  end for
  probability_interval[1] ← 0;
  for i := 2 → NP + 1 do
    probability_interval[i] ← probability_interval[i − 1]
    +probability[i − 1];
  end for
  r ← rand(probability_interval[1], probability_interval[m
+ 1]);
  index ← 1;
  for i := 2 → NP + 1 do
    if probability_interval[i] ≥ r then
      index ← i − 1;
      break
    end if
  end for
  return (population[index], index);
```

Algorithm 5. Mutation_Operator

```
  Require: (k, Π, Q, N, reference_point, shift_point_1,
shift_point_2, max_population)
  for i := 1 → m do
    if rand(0, 1) < Q then
      centre_point ← reference_point[i]
      +H(k)(max_population[i] − reference_point[i])
      +H(k)(shift_point_1[i] − shift_point_2[i]);
      B ← arctan((a_i^R−centre_point/ς(k))
      − arctan((a_i^L − centre_point/ς(k));
      mutated_gene ← reference_point[i] + ς(k)
      · tan(rand(0, 1)/B + arctan((a_i^L
      −center_point)/ς(k));
    else
      centre_point ← reference_point[i];
      B ← arctan((a_i^R − centre_point)/ς(k + N))
      − arctan((a_i^L − centre_point)/ς(k + N));
      mutated_gene ← reference_point[i] + ς(k)
      +N tan((rand(0, 1)/B − arctan((a_i^L
      −centre_point)/ς(k + N)));
    end if
    if mutation_gene > a_i^R then
      mutation_gene ← a_i^R;
    end if
    if mutation_gene < a_i^L then
      mutation_gene ← a_i^L;
    end if
    mutation[i] ← mutation_gene;
  end for
  return mutation;
```

IV. CONVERGENCE OF THE ALGORITHM

It can be demonstrated that the proposed SoFAM algorithm provides the convergence of the sequence of agents in probability to the global maximum with probability one.

Theorem 1.

Let the objective function F be positive in Π. Define the sequence $\zeta(k)$ as follows:

$$\zeta(k) = \left(\frac{1}{k}\right)^{\frac{1}{2n}}. \tag{10}$$

Then, for any point $z \in \Pi$ and any fixed positive real number δ, the sequence of agents generated by the SoFAM algorithm will, with probability one, contain at least one element within the cubic δ-neighborhood of z.

Proof 1.

Let us fix any arbitrary point $z = (z_1, \ldots, z_n) \in \Pi$ and any positive real number $\delta > 0$. These quantities are assumed to remain constant throughout the subsequent analysis and are independent of any other variables or iterative processes.

The cubic δ-neighborhood $O_\delta(z)$ of the point z is giver by $O_\delta(z) = \{y = (y_1, ..., y_n) \colon |y_i - z_i| < \delta, i = \overline{1,n}\}$. According to (8), the probability Q_k for hitting agent z_k into $O_\delta(z)$ with given shift parameters $m_i^k, i = \overline{1,n}$, is expressed as:

$$Q_k = \prod_{i=1}^{n} B_{ik} \int\limits_{z_i-\delta}^{z_i+\delta} \frac{\sigma_{ik}}{(z_i^k - m_i^k)^2 + \sigma_{ik}^2} dz_i^k. \tag{11}$$

Taken into account (9) and the inequality $|arctg(x)| < \pi/2$, we can estimate coefficients B_{ik}:

$$B_{ik} = \frac{1}{arctg(\frac{z_i+\delta-m_i^k}{\sigma_{ik}}) - arctg(\frac{z_i-\delta-m_i^k}{\sigma_{ik}})} > \frac{1}{\pi}. \tag{12}$$

Conditions (8), (9), (10) imply the assessment

$$\zeta(k+N) \le \sigma_{ik} \le \zeta(k) < 1, \tag{13}$$

where N is the preset parameter of a week mutation. Let R be the length of the maximum edge of the hyperparallelepiped Π, $R = \max_{i=\overline{1,n}}(a_i^R - a_i^L)$. Since z^k and m^k belong to Π, the inequalities $|z_i^k - m_i^k| \le R$ holds for $i = 1, 2, ..., n$. Taken into accounts (13), we also get the inequalities:

$$\frac{1}{(z_i^k - m_i^k)^2 + \sigma_{ik}^2} > \frac{1}{R^2+1}. \tag{14}$$

Taken into accounts (10), (11), (12), (14), we can estimate:

$$\begin{aligned} Q_k &> \prod_{i=1}^{n} \frac{1}{\pi} \int\limits_{z_i-\delta}^{z_i+\delta} \frac{\sigma_{ik}}{(z_i^k - m_i^k)^2 + \sigma_{ik}^2} dz_i^k \\ &> \prod_{i=1}^{n} \frac{\sigma_{ik}}{\pi(R^2+1)} \int\limits_{z_i-\delta}^{z_i+\delta} dz_i^k > \frac{1}{\sqrt{k+N}} \left(\frac{2\delta}{(R^2+1)\pi}\right)^n. \end{aligned} \tag{15}$$

The probability P_k^0 of not hitting the agent z_k in $O_\delta(z)$ is expressed through Q_k: $P_k^0 = 1 - Q_k$. Taken into account (15), we can estimate the probability P_M of not getting a single agent into $O_\delta(z)$ during M steps:

$$\begin{aligned} P_M = \prod_{k=1}^{M} P_k^0 &< \prod_{k=1}^{M} \left(1 - \frac{1}{\sqrt{k+N}} \left(\frac{2\delta}{(R^2+1)\pi}\right)^n\right) \\ &< \left(1 - \frac{1}{\sqrt{M+N}} \left(\frac{2\delta}{(R^2+1)\pi}\right)^n\right)^M. \end{aligned} \tag{16}$$

Since

$$\lim_{M\to\infty} \left(1 - \frac{1}{\sqrt{M+N}} \left(\frac{2\delta}{(R^2+1)\pi}\right)^n\right)^M = 0, \tag{17}$$

it follows from the inequality (16) that $P_M \to 0$ when $M \to \infty$. But this also means that the probability of at least one agent belongs to $O_\delta(z)$ tends to one. Due to the arbitrariness of the choice of z and δ, at least one agent generated by SoFAM algorithm will fall into any arbitrarily small neighborhood of any point in Π.

The Theorem 1 is proven.

Theorem 1 means that in the sequence of agents there is an agent that gets into any open subset of the hyperparallelepiped Π with probability one.

This property plays an important role in the theory of evolutionary algorithms [36]. In [37], it was substantiated in the case when a continuous probability distribution is used to generate a new test point at each step of the evolutionary algorithm, the density value of which exceeds a certain limit at any point in the search area. In this case, it is possible to obtain a point in any subset of nonzero measure with nonzero probability at each iteration. As the number of tests increases, the probability of visiting any such subset, including the neighborhood of the global optimum, increases to one. However, the proposed SoFAM algorithm for generating a sequence of test points uses a probability distribution that does not satisfy this case. It can be noted that the exact lower bound of the Cauchy distribution density (8) tends to zero as the parameter $sigma_{ik}$ decreases to zero. Due to this, the specified property for the SoFAM algorithm has to be proven separately, which was done in Theorem 1.

In [37], it was also shown that if this property is satisfied, the sequence of the best found values of the objective function will converge in probability to the value of its global optimum. But from a practical point of view, such convergence is a very weak result. Even the classical Monte Carlo random search method has such convergence. We will show that in the proposed SoFAM method, a much stronger type of convergence takes place - the generated sequence of test points itself converges in probability to the point of global optimum. This means that agents in the SoFAM algorithm are concentrated over time in an arbitrarily small neighborhood of the global maximum of the objective function.

Theorem 2.

Let the objective function F be continuous, strictly positive, and possess a unique global maximum point in Π. Suppose that condition (10) holds. Then, for any fixed positive real number δ, the probability of choosing the agent at the k-th step of the SoFAM algorithm from the cubic δ-neighborhood

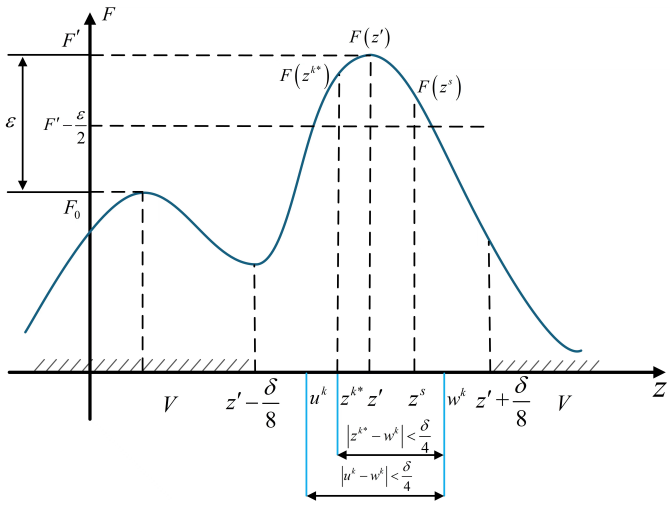

Fig. 1. Geometric illustration of the constructions used in the proof of Theorem 2 for the case n=1

of the global maximum point tends to one with probability one when $k \to \infty$.

Proof 2.

Let $z' = (z'_1, ..., z'_n)$ be the single global maximum point of the function $F(z)$ in Π, $F' = F(z')$, δ be an arbitrarily small positive number. Consider the cubic δ-neighborhood $O_\delta(z')$ and $\delta/8$-neighborhood $O_{\delta/8}(z')$ of the point z'. Define V as the complement of $O_{\delta/8}(z')$ in Π; $F_0 = \max_{z \in V} F(z)$ (see Fig. 1).

Obviously, $F_0 < F'$. Let us introduce the notation $\varepsilon = F' - F_0$, $\varepsilon > 0$. Since function F is continues, there exists an open subset of $O_{\delta/8}(z')$ where $F(z) > F' - \varepsilon/2$ (see also Fig. 1). Then, according to Theorem 1, with probability one, there exists an agent z^s in this subset, i.e. $z^s \in O_{\delta/8}(z')$ and $F(z^s) > F' - \varepsilon/2 > F_0$.

Consider the step k greater than s. Then according to (2), $F(z^{k*}) \geq F(z^s) > F' - \varepsilon/2$, hence $z^{k*} \in O_{\delta/8}$ (see also Fig. 1).

If z^j is an agent of the population from V, we can estimate probabilities (3) of its selection as the reference agent w^k or additional selected agents v^k, u^k at the k-th step:

$$p_j^k = \frac{\left(F(z^j)\right)^{\phi(k)}}{\sum_{l=1}^{k-1} \mu(l, k-1)\left(F(z^l)\right)^{\phi(k)}} < \frac{(F_0)^{\phi(k)}}{\left(F(z^{k*})\right)^{\phi(k)}} < \left(\frac{F' - \varepsilon}{F' - \frac{\varepsilon}{2}}\right)^{\phi(k)}. \tag{18}$$

Since the left-hand side of the inequality (18) tends to zero, this probability tends to zero as the number of steps increases indefinitely. Consequently, the probability of selection of agents w^k, v^k, u^k from the neighborhood $O_{\delta/8}(z')$ tends to one as $k \to \infty$. Then the probability of choosing all three of these agents at once from this neighborhood also tends to one. In this case $|w_i^k - z'_i| < \delta/8$, $|v_i^k - z'_i| < \delta/8$, $|u_i^k - z'_i| < \delta/8$ and $|w_i^k - u_i^k| \leq |w_i^k - z'_i| + |u_i^k - z'_i| < \delta/4$. Moreover, $|z_i^{k*} - z'_i| < \delta/8$; $|z_i^{k*} - w_i^k| \leq |z_i^{k*} - z'_i| + |w_i^k - z'_i| < \delta/4$ (see also Fig. 1).

Then we can estimate the shift parameter of i-th gene: for the strong mutation:

$$|m_i^k - z'_i| \leq |w_i^k - z'_i| + H(k)|z_i^{k*} - w_i^k| + H(k)|v_i^k - u_i^k| < \frac{5\delta}{8}; \tag{19}$$

for the weak mutation:

$$|m_i^k - z_i'^k| < \frac{\delta}{8} < \frac{5\delta}{8}. \tag{20}$$

Using the estimation (12) for B_{ik}, we can estimate the probability $Q_k(z')$ of belonging the agent z^k to the neighborhood $O_\delta(z')$ as follows:

$$Q_k(z') = \prod_{i=1}^{n} B_{ik} \int_{z'_i - \delta}^{z'_i + \delta} \frac{\sigma_{ik}}{(z_i^k - m_i^k)^2 + \sigma_{ik}^2} dz_i^k > \prod_{i=1}^{n} \frac{1}{\pi}\left(arctg\left(\frac{z'_i + \delta - m_i^k}{\sigma_{ik}}\right) - arctg\left(\frac{z'_i - \delta - m_i^k}{\sigma_{ik}}\right)\right).$$

From (19), (20) we have $z'_i + \delta - m_i^k > \delta - \frac{5\delta}{8} > 0$. Since $\sigma_{ik} \to 0$ as $k \to \infty$, then $arctg((z'_i + \delta - m_i^k)/\sigma_{ik}) \to \pi/2$.

Similarly, $z_i' - \delta - m_i^k < -\delta + \frac{5\delta}{8} < 0$, then $arctg((z'_i - \delta - m_i^k)/\sigma_{ik}) \to -\pi/2$. In this case $Q_k(z') \to 1$.

The Theorem 2 is proven.

An important condition for applying the SoFAM algorithm is the positivity of the objective function F. If this condition is not met, then formula (3) can give a negative result, which cannot be used as a probability of choosing an agent. It should also be noted that the algorithm is designed to solve the problem of finding a global maximum point. However, in the case where an objective function J has an arbitrary sign and is required to find its global minimum, it is easy to make simple transformations of J, in which the equivalent problem will already meet the necessary requirements for applying the SoFA algorithm.

One of the simplest transformations of this type is the transition from the original objective function J to a new function F according to the following formula

$$F_1 = exp(-J). \tag{21}$$

Obviously, the function (21) is strictly positive everywhere and its maximum point coincides with the minimum point of the original function J.

Such a transformation is not the only one. For example, the following transformation of the objective function can be used

$$F_2 = A/(C + exp(J)), \tag{22}$$

where A and C are arbitrary positive constants. This transformation coincides with the previous one (21) for $A = 1$, $C = 0$. Fig. 2 shows the original function $J = -sin(2(z-2))/(z-2)$ with an extension of the definition by continuity at the point $z = 2$ ($J(2) = -2$), along with two transformed functions according (21) and (22) with $A = 1$, $C = 1$.

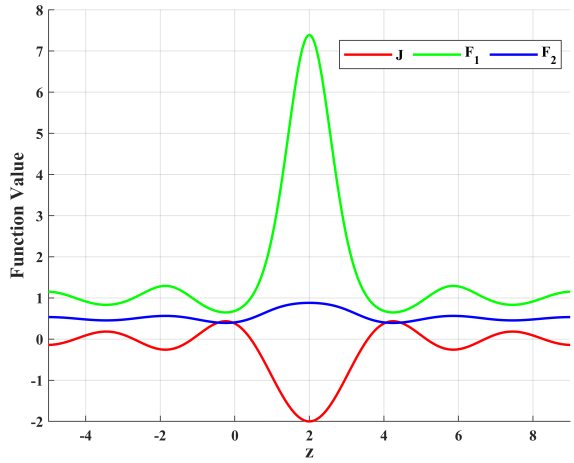

Fig. 2. Geometric illustration of the objective function transformations.

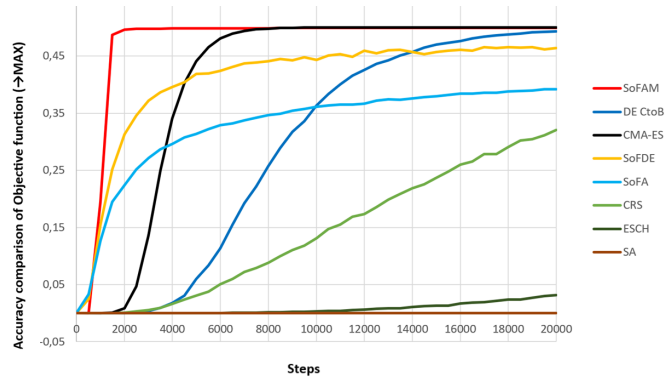

Fig. 3. Comparison of the accuracy of algorithms. For testing, 100 functions of the class GKLS were randomly generated using the transformation (31) of $J(z)$. The search space dimension was n=50, the initial population size was m=100, the number of local extrema was 1000, and half the side of Π was 10. We used each algorithm to optimize each function with a different maximum number of iterations (from 0 to 20,000 with a step of 500; the result on the initial population corresponded to zero). The figure shows the average result for 100 random functions.

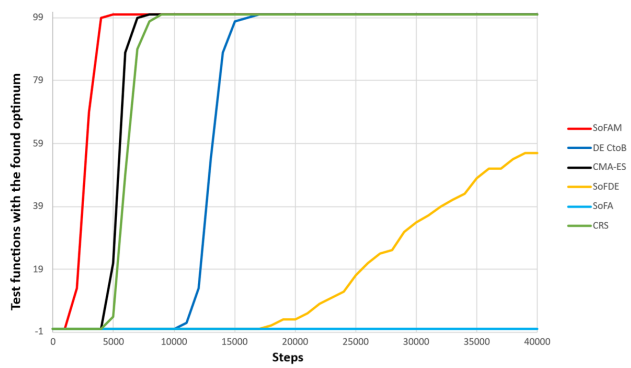

Fig. 4. Results of the competition of algorithms in the GKLS class. The vertical axis shows the percentage of test functions for which the algorithm could determine the real optimum. It is considered that the actual optimum is found if the Euclidean distance of the point obtained by the algorithm differs from the actual global maximum by less than 0.15. For testing, 100 functions were randomly generated using the transformation (31) of $J(z)$. The search space dimension n=25, the initial population size m=100, the number of local extrema is 100, and half the side of the Π is 5. We use each algorithm to optimize each function with a different maximum number of iterations (from 0 to 40,000 with a step of 1,000; the result on the initial population corresponds to zero).

V. EXPERIMENTAL RESULTS

A. Performance Comparison

Algorithm SoFAM was compared with other well-established global optimization algorithms on relevant test functions to evaluate the practical significance and performance of the algorithm. Algorithms of the same class were selected for comparison: basic SoFA algorithm, controlled random search with local mutations CRS [38], evolutionary algorithm ESCH [39], differential evolution (current to the best) DE CtoB [40], covariance matrix adaptation evolution strategy (CMA-ES) [42] and the well-known annealing algorithm SA [41]. In addition, the previous accelerated modification of the SoFA algorithm, the SoFDE algorithm, was considered [31].

To reduce the impact of the choice of a specific test function on the comparison of the performance of global optimization algorithms, a wide class of non-differentiable randomly generated test functions, known as GKLS (ND type), was used [43].

In this class, we can specify the location and number of local maximum points and the most significant values of the generated functions in advance. Using such a class of test functions allows us to reduce the total number of experiments required to evaluate the algorithm's performance adequately.

The study was conducted on functions defined in a 50-dimensional hypercube $\Pi = \{z = (z_1; ...z_{50}) : -10 \leq z_i \leq 10, i = \overline{1, 50}\}$. The number of local extremum points was set to 1000, the position of the global extremum point was generated randomly, and the value of the global maximum was set to 0.5.

A set of 100 GKLS functions was randomly generated, followed by 100 independent runs of each optimization algorithm. Here we used the following values of pre-set parameters of SoFAM algorithm: $\phi(k) = (k/1000 + 1)^{0.5}$, N=20000, G=0.7, $H(k) = 0.5(1 - \tanh(k - 10000))$, γ=0.01. The settings were selected empirically to ensure maximum observed performance. The comparison results are illustrated in Fig. 3, where the horizontal axis represents the number of iterations, and the vertical axis displays the achieved average maximum value of the objective function. The results indicate that the proposed algorithm outperforms established alternatives, including the previous modification.

With a given number of iterations, only the proposed algorithm SoFAM and SMA-ES achieved the actual global maximum of the objective function with an error of less than 0.1%, showing the highest accuracy. The closest accuracy algorithm, SoFDE, decreased accuracy by 7% compared to the leader. For the remaining algorithms, the accuracy dropped from 24% (DE CtoB) to zero (SA) compared to the proposed SoFAM.

The optimization algorithms were also compared by their ability to get into a small neighborhood of a fixed global maximum after a certain number of steps. The following criterion for optimization success was used: the Euclidean distance between the true point of the global maximum and its approximation should be less than a certain threshold value of

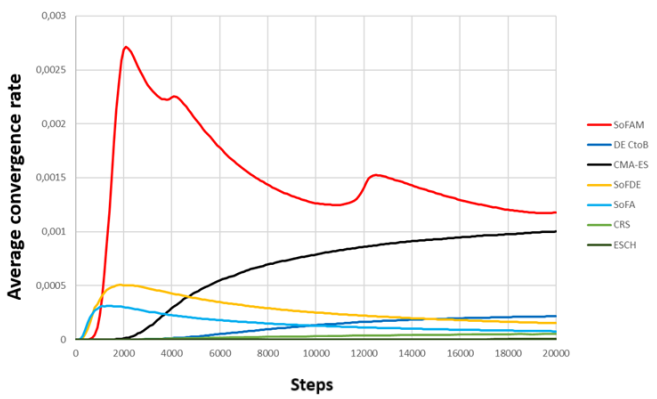

Fig. 5. Estimations of the average convergence rate in the GKLS class. The vertical axis shows the average convergence rate of algorithms.

0.15. In the hypercube, 100 samples of GKLS functions were generated with the number of extremum points equal to 100. Then, 10 runs of each algorithm were performed. The results of the experiment are shown in Fig. 4. The horizontal axis shows the number of steps, and the vertical axis shows the percentage of fulfillment of the success criterion for different algorithms depending on the number of steps. It can be seen that the new algorithm outperforms its previous analogs by this criterion as well.

The experiments conducted to study the rate of the algorithms also confirmed the advantage of SoFAM: the proposed algorithm reached the specified small 0.1% neighborhood of the global maximum faster than all in 3000 iterations. For comparison, the closest in terms of performance, CMA-ES and CRS, required three times more iterations (9000), while the other algorithms showed even lower performance.

We also estimated the average convergence rate of the algorithms in the GKLS class of functions, determined by the formula [44]

$$R_k = 1 - |\frac{F(z') - F(z^{k*})}{F(z') - F(z^{1*})}|^{1/k}$$

.

The method given in [45] was used to calculate it. The calculation results are shown in Fig. 5. They show that the average convergence rate in the GKLS class for the SoFAM algorithm is higher than for all other comparison algorithms considered.

Comparative experiment results demonstrate the potential effectiveness of the new algorithm for solving various optimization problems.

B. Complexity Analysis of The SoFAM algorithm

We will consider the algorithmic complexity of the proposed SoFAM algorithm according to the worst-case metric.

The time complexity of the SoFAM algorithm depends on several parameters: n (the dimension of the search space), m (the size of the initial population), and k (the current step of the algorithm).

Before starting the algorithm, it is essential to generate the initial population of agents, with the complexity of this procedure being $O(m \times n)$. The overhead incurred from the computation of the objective function will be addressed subsequently. Upon completion of the initialization phase, we proceed to analyze the time complexity associated with a single iteration of the algorithm.

The first step in each iteration involves selecting a test reference agent and auxiliary agents to shift the mutation using the corresponding formula. In the worst case, where the population contains the maximum number of individuals, this procedure requires $O(m)$ time. Subsequently, the creation of an agent using the anisotropic mutation operator is necessary, which incurs an additional cost of $O(n)$. Therefore, the time complexity of an iteration of the SoFAM algorithm is $O(m) + O(n)$.

In the algorithm presented here, the complexity associated with a specific iteration does not depend on its iteration number. Consequently, for k iterations, the total time complexity is $O(m \times n) + O(m \times k)$. Therefore, the overall time complexity of the algorithm, in terms of parameters, is $O(m \times n) + O(m \times k) + O(n \times k)$. Given that both m and n are treated as constants, the asymptotic time complexity of the algorithm is primarily influenced by k. As a result, we obtain: $O(m \times n) + O(m \times k) + O(n \times k) = O(1) + O(k) + O(k) = O(k)$. Thus, the time complexity of the algorithm can be succinctly expressed as $O(k)$, indicating that the SoFAM algorithm exhibits linear time complexity with respect to the number of iterations.

C. Application Example for Optimal Control of UAVs

To study the applicability of the developed SoFAM method to solving practical engineering problems, we applied this method to determine the optimal values of the transfer functions of the autopilot of a medium altitude long endurance UAV SR200 design of Xi'an Supersonic Aviation Technology (https://satuav.com/glider-drone-runway-drone/medium-altitude-long-endurance-male-uav.html). We considered the turn maneuver in turbulent gusts of wind [46].

The mathematical model describing the movement of the aircraft in gusts of wind has the form:

$$\dot{Y}_1 = -k_\beta(Y_1) + \alpha_0 Y_2 + Y_3 + k_\gamma Y_4, \quad (23)$$

$$\dot{Y}_2 = -a_0(Y_1) - a_1 Y_2 - a_2\big(Y_3 - \omega_{y*}\big) + a_3 Y_6 - l_\psi \omega_{y*}, \quad (24)$$

$$\dot{Y}_3 = -a_4(Y_1) - a_5 Y_2 - a_6 Y_3, \quad (25)$$

$$\dot{Y}_4 = Y_2, \quad (26)$$

$$\dot{Y}_5 = Y_3 - \omega_{y^*}, \quad (27)$$

where $(a_0, a_1, a_2, a_3, a_4, a_5, a_6)$ donate autopilot parameters; $\omega_{y*} = 2$grad/s indicates the specified target value of angular velocity during turn; $(Y_1, Y_2, Y_3, Y_4, Y_5)$ represent state variables, they correspond to slip angle β, angular velocity ω_x, angular velocity ω_y, roll angle γ, and deviation of the UAV angular velocity from the target value, respectively. $k_\gamma = 0.0833$(1/s), $k_\beta = 0.1946$(1/s), $\alpha_0 = 3.9° = 0.0680678$(rad), $l_\psi = 1.742$(1/s) indicate the fixed parameters of the UAV.

Characteristics of the SR200:
- max take off weight: m=320 kg;
- wingspan: l=8,1 m;
- max flight height H_{max}=5000 m;
- max flight speed V_{max}=190 km/h;
- aircraft wing area S=10m^2;
- trim angle of attack α_0=3.9°;
- lateral force coefficient C_z^β=0,15.

Flight conditions:
- flight altitude during turn – H_0=1000 m;
- air density at flight altitude ρ_0=1,1117 kg/m^3;
- wind speed at flight altitude - 50 m/s;
- true air speed V_0=111 m/s.

Turbulence characteristics:
- turbulence scale L_{W_x}=310 m;
- pulsation intensity σ_{W_x}=2.8 m/sec.

Based on the formulated conditions, the dynamic coefficients of the UAV have the following values:
k_γ=0,0833 (1/s),
k_β=0,1946 (1/s),
α_0=3,9°=0,0680678 rad.,
l_ψ=1,742 (1/s).

Calculating the autopilot parameters $a_0 - a_6$ is necessary to minimize the deviation of the actual angular velocity $\omega_y = Y_3$ from the specified one. In other words, it is necessary to ensure that the following criterion:

$$J = \int_0^{t_k} (Y_3^2 - 2Y_3\omega_{y*} + \omega_{y*}^2)dt \tag{28}$$

reaches its minimum value.

The following constraints are imposed on the autopilot parameters:

- a_0 and a_5 are control functions whose values belong to the following segments:

$$\begin{cases} a_0(t) & \in [0,2] \\ a_5(t) & \in [0,3] \end{cases}$$

- the remaining control parameters are constants from the following segments:

$$\begin{cases} a_1 & \in [4,12] \\ a_2 & \in [-200,1.74] \\ a_3 & \in [0,176.54] \\ a_4 & \in [7,21] \\ a_6 & \in [0,8]. \end{cases}$$

In accordance with the classical theory of optimal control, the following conjugate functions are introduced: $\psi_1, \psi_2, \psi_3, \psi_4, \psi_5$. They satisfy the adjoint system of equations:

$$\begin{cases} \dot{\psi}_1 & = k_\beta\psi_1 + a_0\psi_2 + a_4(t)\psi_3 \\ \dot{\psi}_2 & = -a_0\psi_1 + a_1\psi_2 + a_5\psi_3 - \psi_4 \\ \dot{\psi}_3 & = -\psi_1 + a_2\psi_2 + a_6\psi_3 - \psi_5 - 2Y_3 + 2\omega_{y*} \\ \dot{\psi}_4 & = -k_\gamma\psi_1 \\ \dot{\psi}_5 & = -a_3\psi_2 \end{cases}$$

and the conditions of transversality:

$$\psi_1(t_k) = \psi_2(t_k) = \psi_3(t_k) = \psi_4(t_k) = \psi_5(t_k) = 0.$$

The Hamilton function has the following form:

$$\begin{aligned} H = &- k_\beta\psi_1Y_1 - a_0\psi_2Y_1 - a_4(t)\psi_3Y_1 + \alpha_0\psi_1Y_2 \\ &- a_1\psi_2Y_2 - a_5\psi_3Y_2 + \psi_4Y_2 + \psi_1Y_3 - a_2\psi_2Y_3 \\ &- a_6\psi_3Y_3 + \psi_5Y_3 + k_\gamma\psi_1Y_4 + a_3\psi_2Y_5. \end{aligned}$$

According to the maximum principle of Pontryagin [3, 47], optimal control must satisfy the following necessary conditions:

$$a_0 = \begin{cases} 0, & \text{if} \psi_2Y_1 > 0 \\ 2, & \text{if} \psi_2Y_1 < 0 \end{cases} \tag{29}$$

$$a_5 = \begin{cases} 0, & \text{if } \psi_3Y_2 > 0 \\ 3, & \text{if } \psi_3Y_2 < 0 \end{cases} \tag{30}$$

where functions Y_1, Y_2, ψ_2, ψ_3 are the components of the solutions of the corresponding controlled and adjoint systems under optimal control.

In addition, optimal control can have singular regimes. A singular optimal regime is an optimal control for which, over a certain period of time, the first and second derivatives of the Hamiltonian function H with respect to the control variable a simultaneously vanish: $\partial H/\partial a = 0$, $\partial^2 H/\partial a^2 = 0$ [8, 9]. If, for example, the Hamiltonian is a linear function of this variable: $H = \psi a + c$, and the linear coefficient ψ of this function vanishes over a certain time interval, then the application of the classical principle minimum does not allow us to find a solution, since maximizing the Hamiltonian H with respect to the control variable a does not provide any useful information. Here, it becomes necessary to construct a solution based on necessary or sufficient conditions of the second or higher order [48–50]. Optimal control in this time interval is called a special optimal regime. An example of a problem with special control is optimization of the thrust of a missile flying at a constant altitude and which is launched at low speed, considered in [51].

In our problem, it is possible a singular optimal regime on intervals where the equalities $\psi_2Y_1 = 0$ or $\psi_3Y_2 = 0$ holds. In this case, it is impossible to construct optimal control directly from these necessary optimality conditions since it is only possible to determine the signs of the functions Y_1, Y_2, ψ_2, ψ_3 in advance by knowing the optimal control. It further complicates the solution.

To facilitate the numerical solution, the control time segment was divided into T intervals $[t_{i-1}, t_i], i = \overline{1,T}$. For simplicity in calculations, we adopted a constant interval length throughout the entire control period. On each i-th interval, we considered the control constant: $a_0(t) = a_i^0, a_5(t) = a_i^5, t \in [t_{i-1}, t_i], i = \overline{1,T}$, that is, we formed a piecewise constant approximation of the control. Next, the controlled system was integrated using the Runge-Kutta algorithm for the selected constant values, and the value of the quality criterion was calculated. The problem was to find such piecewise constant functions for which the quality criterion reaches its lowest

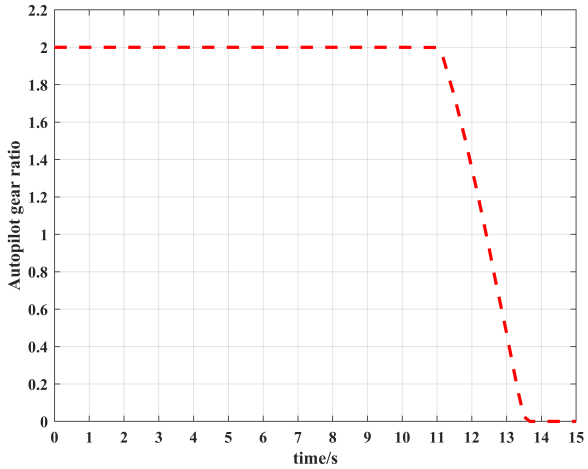

Fig. 6. Autopilot control function with switching.

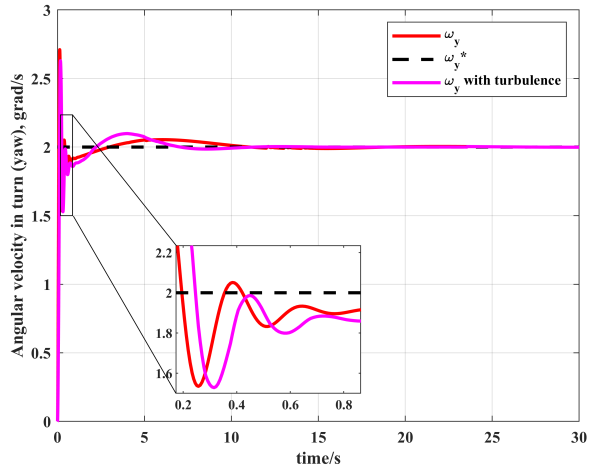

Fig. 7. Transient process of angular velocity stabilization.

value. Thus, we reduced the original optimal control problem to a multidimensional optimization problem.

The proposed optimization algorithm, SoFAM, was used to solve this problem. Note that SoFAM was initially developed to solve the maximum problem, and the applied problem being solved requires finding the minimum. We used the following transformation of the original criterion J to solve the difficulty that arose:

$$F = \frac{1}{1 + \exp(J)}. \tag{31}$$

It is easy to see that the functional J minimum corresponds to the maximum of functional F. In addition, the objective function F becomes strictly positive, which ensures the fulfillment of the second condition of SoFAM applicability.

For the numerical experiment, we set $T = 90$ and solved the resulting optimization problem in a 185-dimensional space.

The numerical experiment showed that the objective function has the best value for the constant control $a_5(t) = 3$. The control function $a_0(t)$ is graphically represented in Fig. 6.

The remaining constant control parameters are respectively equal to: $a_1 = 5.98863558$, $a_2 = -200$, $a_3 = 176.54$, $a_4 = 7$, $a_6 = 8$.

With this control we obtain the following value of the objective function 0.2081. It is less than the best value when all control variables are constant.

This control ensured a stable transition process with an exit to a stable mode. The actual angular velocity Y_3 is stabilized at a given value ω_{y*} (Fig. 7).

However, it is known that the numerical solution may not correspond to the true one due to the inaccuracy of the approximations and the accumulation of computational errors. Therefore, the obtained solution was checked based on the previously established necessary optimality condition. To carry out the check, the controlled and conjugate systems with the obtained numerically optimal control were integrated, then the zeros of the products $\psi_2 Y_1$ and $\psi_3 Y_2$ were found, which were then compared necessary conditions.

From Fig. 7, it is evident that the value of the phase variable Y_3 stabilizes at 4 starting from the 3rd second. But then the solution of the adjoint system becomes trivial when approaching the final moment. Due to this, the maximum principle does not give a certain value of optimal control on this interval, i.e. a special regime takes place.

It is known that the numerical solution may not correspond to the true one due to the inaccuracy of the approximations and the accumulation of computational errors. Therefore, the obtained solution was checked based on the previously established necessary optimality condition. To carry out the check, the controlled and conjugate systems with the obtained numerically optimal control were integrated, then the zeros of the products $\psi_2 Y_1$ and $\psi_3 Y_2$ were found, which were then compared necessary conditions.

The comparison showed fulfillment of the necessary optimality conditions for the constructed control.

To evaluate the efficiency of new approach for solving the stated problem, it was solved in the class of constant controls by different optimization algorithms.The algorithms were carefully selected for a rigorous comparison: differential evolution (current to the best) DE CtoB, annealing SA, basic SoFA and the SoFDE algorithm. Each algorithm completed 3000 iterations. Fig. 8 shows the comparison results.

The SA and SoFDE algorithms get stuck in a local minimum and give a false solution that is not a global optimum. Differential evolution and SoFAM find a better solution, but SoFAM works faster. Thus, within the framework of the model under study, the SoFAM algorithm is preferable.

The result obtained indicates the advantages of the new algorithm. This example shows the prospects of using the new SoFAM algorithm to solve applied optimal control problems.

VI. Discussion

In this paper, a new evolutionary algorithm of global optimization SoFAM was proposed. It is an upgrae of the existing SoFA algorithm. The most important advantage of this algorithm, as well as the SoFA algorithm, compared to most heuristic evolutionary algorithms is the proven convergence.

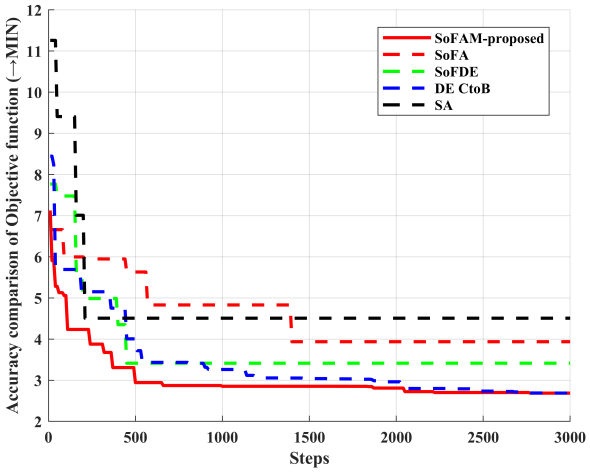

Fig. 8. Comparison of algorithms for UAVs control.

In the proposed algorithm, unlike all other evolutionary algorithms, the population size changes according to the dynamics of the Volterra-Lotka model. The mutation operator acts differently than in the SoFA – it is based on the Cauchy distribution and shifts the distribution center (location parameter) like the mutation operator in differential evolution. However, unlike the latter, the shift parameter (regulating its length) gradually decreases with an increase in the number of steps. This enhances the effect of a gradual decrease in the scaling parameter (variance in the normal distribution) and accelerates the approach to the global maximum point in the final stages of the process. Instead of the crossover operator, which is typical for evolutionary algorithms, here, like in the SoFA algorithm, mutations of two different types are used - strong and weak. The purpose of introducing different types of mutations is to imitate the crossover effect, but in such a way as to preserve the main advantage of the algorithm - its convergence. At the same time, the proposed SoFAM algorithm has significant differences from the differential evolution algorithm. First, this difference lies in the choice of the reference agent (parent)for the formation of the next mutation. According to the differential evolution algorithm, this agent is selected from the population points with equal probability. In the SoFAM algorithm, this selection is also random, but the distribution of selection probabilities corresponds to the dynamic processes of competition and selection in the Volterra-Lotka model. This allows us to select reference agents taking into account the values of the objective function, which, on the one hand, increases the efficiency of the algorithm and, on the other hand, is an important factor ensuring convergence. In addition, if we draw a parallel between the two algorithms, we can see that performing the crossover operation in differential evolution is actually equivalent to only shifting the center of the mutation distribution (location parameter) of each gene in the SoFAM algorithm. In the SoFAM algorithm, after performing this transformation, an additional random scatter relative to the shifted center (location) is used according to the Cauchy distribution, which has no analogue in differential evolution. This provides an additional examination of the search space and has a positive effect on productivity. The intensity of this scatter (scale parameter) is not constant, but gradually decreases to zero, due to which the convergence of the proposed algorithm is achieved.

Note that the transition to the Cauchy distribution is not a trivial technique since it requires an independent justification for convergence [31, 52]. Here, a different formula for the dependence of the scaling parameter on the number of steps (10) was obtained for the convergence of the algorithm than for the decrease in variance in the SoFA algorithm. Note also that, unlike the SoFA algorithm, this formula does not depend on the size of the search area (the length of the edges of the hyperparallelepiped Π).

For practical use of the algorithm, it is important to estimate its performance in the class of relevant functions, for example, CEC-2017, CEC-2022, GKLS, etc. [43, 53–55]. In this case, a certain class of functions with a priori properties and parameters is fixed, and a sample of representatives of this class is formed. Then, the average number of steps that must be taken to reach the given neighborhood of the global extremum is empirically established (see, for example, [56]). Here, the GKLS class of functions was used [43]. Unlike other widely used fixed sets of test functions (CEC-2017, CEC-2022, etc.), this class is infinite, and the sample of functions is formed randomly, which excludes the possibility of using any additional private information about the target function to adjust the algorithm, except for the known properties of the class as a whole. The result shown on such a sample is closest to the objective characteristic of the algorithm. The data obtained on a sample of 100 GKLS functions are shown in Fig. 4. It is evident from this that to reach the neighborhood of the global maximum with a radius of 0.15 in a hypercube with an edge length of 10 in a 25-dimensional Euclidean space, the SoFAM algorithm requires no more than 5000 steps. The conducted study also shows that the proposed SoFAM algorithm has a higher average convergence rate on functions of the selected class compared to the considered traditional (metaheuristic) algorithms. One can also consider exhaustive optimization strategies (checking all feasible solutions), for example, the enumeration method. In order for the distance from a grid point to the global extremum to be less than 0.15 in a hypercube with an edge length of 10 in 25-dimensional Euclidean space, it is necessary to construct a grid of $3 \cdot 10^{55}$ points. This value is incommensurably greater than the 5000 steps required by the SoFAM algorithm.

The SoFAM algorithm can be applied only in the case of a continuous objective function. This is the main limitation for its use.

Another condition for applying this algorithm is that the objective function is positive. If the objective function allows negative values, then the direct application of the SoFAM algorithm is impossible, and a preliminary transformation of the objective function is required to ensure the desired condition. Such a transformation is always possible and in various ways, for example (21), (22) One of them, based on formula (31), was used in this study to solve an applied optimal control problem.

The next condition for applying the algorithm is the presence of one global extremum in the objective function. However, Theorem 1 is also true when the objective function has several global maximum points. This theorem states that at least one test point generated by the algorithm in a sufficiently large number of steps falls into an arbitrarily small neighborhood of any point in the search area, including any global maximum point. Thus, the algorithm potentially allows us to find arbitrarily accurate approximations of all global maximum points. However, to use this useful property of the algorithm, it is necessary to change the output information. It is impossible to limit ourselves to only the values z^{j*} and $F(z^{j*})$. Due to the fact that the global maximum points are determined approximately, the values of the objective function in them may differ slightly and be less than $F(z^{j*})$. Therefore, possible solutions will be test points z^j that satisfy the condition $F(z^j) > F(z^{j*}) - \theta$, where $\theta > 0$ is a preset parameter of the accuracy of determining the maximum value of the objective function.

At the same time, the new algorithm is focused primarily on solving optimal control problems. In this class of practically significant problems, optimal control is often the only thing that justifies the use of this algorithm. As practice shows, the SoFAM algorithm, like many other search algorithms, works faster in the case of a relatively flat graph of the objective function and significantly slower in the presence of numerous elongated peaks in the objective function. However, in optimal control problems, a picture of a relatively calm change in the values of the objective function is also more often observed, which makes the proposed algorithm effective in solving such problems.

The modifications introduced and the correct use of their capabilities made it possible to ensure a higher performance of the SoFAM algorithm in practice, including compared to the basic SoFA algorithm and the differential evolution. It was demonstrated in a number of experiments with test GKLS functions.

After checking with the GKLS test functions, the proposed algorithm was applied to solve the problem of optimal control of a UAV during a turn. The presence of a special control mode characterizes this problem. In addition, the problem was a combination, i.e. we simultaneously found optimal values for several constants and optimal modes for control parameters changing over time. The application of the new algorithm made it possible to find effective control that ensured the transition to steady-state mode with an acceptable overshoot in just three seconds (out of the 15 seconds initially allocated for control). At the same time, the deviation from the program trajectory in angular velocity was less than 0.01%. Then, the optimal control problem was solved for more complex operating conditions of the aircraft, considering movement in a strong, gusty crosswind (turbulence). The model contained 26 differential equations. This problem was characterized by the long time to calculate the value of the objective function for each trial control. Calculations showed that constant control would be optimal here. The implemented control strategy guided the UAV to achieve stable operation in 10 seconds.

We compared the efficiency of the new algorithm with some known algorithms for the problem of optimal control of UAVs under turbulence. We used the base SoFA algorithm and its modification SoFDE, differential evolution DE CtoB, and annealing algorithm SA. The new algorithm works faster and more accurately than the analogues considered. When solving the problem for the minimum, the optimal value found by our algorithm was 2.686 while achieving the neighborhood of 1% in 640 iterations. The DE CtoB algorithm showed the same accuracy in 2030 iterations, and the value of the fitness function at the minimum point found in DE CtoB is 2.688. The other algorithms did not reach the global minimum at all.

Thus, the new SoFAM algorithm is more accurate and faster than its analogs, and researchers can successfully apply it to solving problems of optimal control of complex technical systems.

VII. Conclusion

In this paper, a new evolutionary algorithm for global optimization is proposed, for which it was possible to prove the convergence in probability of the generated sequence of test points to the global extremum point. This result is very important in the theory of numerical optimization algorithms, since it opens up the possibility of developing and improving algorithms with guaranteed convergence.

It should be noted that the new algorithm, like other evolutionary algorithms, is based on the ideas of biological evolution put forward by Ch. Darwin. However, this algorithm uses deeper and more detailed patterns of known biological processes. In particular, it uses ideas about the decrease in the intensity of mutagenesis over time, different mutation intensities for different genes, and changing distribution centers of mutations. In addition, when selecting parents for the production of offspring, formulas that correspond to the competition of species in the Volterra-Lotka population model are used.

The comparative experiments conducted showed the advantage of our algorithm both in accuracy and in convergence rate.

We applied the SoFAM algorithm to calculate the optimal control of the UAV turn for both turbulence-free and turbulent conditions. In both cases, the algorithm allowed us to find an effective control that ensures that the UAV quickly enters a stable mode.

Thus, it can be concluded that the conducted studies show high theoretical and practical significance of the SoFAM algorithm presented.

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